

Winterborn Restock System

Development Brief — Document 3: Reconciliation with the 22 Aug 2026 CEO Voice Note

Internal document for the build team. Not client-facing.

Reconciles: `Winterborn-Dev-Brief-Document-2.pdf` against `Latest-CEO-Transcript-Summary.md`

Authority order set by the project owner, 24 Aug 2026: where the two conflict, the voice note wins. Where Document 2 is silent or the two agree, Document 2 stands. Nothing here overrides the technical facts established by real client data (the Square and Sortly exports) or the proven safety design in the catalog migration decision record.

This is a scope document, mirroring Document 2's own role. It answers *what* changes and *why*. For implementation detail see `docs/superpowers/specs/2026-08-19-winterborn-restock-system-design.md` and the six build plans under `docs/superpowers/plans/`.

1. What has not changed

The voice note independently re-derives the same core architecture Document 2 already specified — in places, almost word for word. This is a strong signal the underlying design was right, not a reason to redesign it. The following stand exactly as written in Document 2:

- **Inventory is a derivation, never a stored number.** Voice note §13: *"The platform must therefore maintain a reliable inventory movement ledger rather than simply maintaining one global stock number."* This is our exact architecture, described independently by the client.
- **Location-level inventory, not company-wide totals.** Voice note §43 states this as "the most important design principle." It already was ours.
- **Two-tier colour model** — till sees family only ("Blue"), warehouse holds full detail. Voice note §5.2, §9 match this design exactly, down to the same worked example (a range of blue shades collapsing to one till option).
- **PWA, not native apps.** Voice note §12 confirms this explicitly.
- **No inventory writes into Square in v1;** Square remains the sales record, read via webhook and reconciliation poll. Voice note §15 describes exactly this flow.
- **Catalog restructuring protocol** (§4 of Document 2) — prototype first, read-modify-write, preserve `item_id`, the Sept 12 freeze. Untouched. Already built and proven against Square sandbox; see the migration decision record.
- **No colour-level demand prediction this season.** 2025 data carries colour on 4.8% of revenue. This stays forbidden regardless of the voice note (see §4 below on why Smart Shipments does not conflict with this).
- **All six catalog write-safety guards** — backup-before-write, allowlist, no-delete, stop-on-failure, rollback, preflight. Untouched.

2. Reconciliation summary

Area	Document 2	Voice note	Decision
Warehouse intake / Sortly replacement	Stage 2, ~12 hrs, late October	§10–11: detailed receiving workflow, camera capture, described as core to daily operation. Independently, the project owner was asked directly by the CEO to review Sortly "today" and "replace this tool."	PULLED TO v1
Automated low-stock alerting	Stage 1 explicit: <i>"thresholds exist but alerts are manual-review only... Stage 2 turns automated alerts on."</i>	§16, §33: system should proactively flag without a human noticing first.	PULLED TO v1 (detection layer — see §4.2)
Restock quantity suggestion	Not specified; requests carry only the requested quantity	§18, §20–23 "Smart Shipments": recommend a quantity from historical velocity, current stock, remaining season days	PULLED TO v1 — style level only, see §4.3
Overallocation across markets	Not specified	§19: cap a request when fulfilling it in full would starve another location	PULLED TO v1
Shipment consolidation	Not specified	§24: bundle nearby restock needs into one dispatch run	PULLED TO v1
Roles	Four: Owner, Warehouse, Market Manager, Operator	§26: five, splitting Warehouse into Manager/Operator tiers, adding a Square-only Sales Operator	PULLED TO v1
Dashboard status	Low-stock list, decision queue	§17, §25: explicit Healthy / Low / Critical / Out-of-Stock zones	PULLED TO v1 — labelling on existing data
QR box tracking	Core to dispatch: one label per box, scan at the warehouse door	§14 explicit: <i>"QR-based box tracking is optional... mandatory QR scanning is unnecessary"</i> unless discrepancies later prove it's needed	SOFTENED — see §5
Physical warehouse tiers	Hard constraint: <i>"all discipline lives at ONE warehouse"</i>	§6–7: describes an optional Chicago → market-level sub-warehouse → booth chain, hedged as <i>"not every market"</i>	NEEDS ONE FACT — see §6.1

		<i>necessarily requires a separate warehouse"</i>	
AI image recognition at intake	Not specified	§11, the voice note's own words: <i>"may later help"</i>	STAYS STAGE 2 — deferred by the source itself
New-market cold start (weather, demographics, population)	Not specified	§22, listed as a possible future input for markets with no history	STAYS STAGE 2 — no data pipeline exists; premature
Write-off flow, season-close reconciliation	Stage 2	Not mentioned	UNCHANGED

3. Why these pull forward safely

None of the items below need new architecture. Each is additive to what already exists, verified against the current codebase rather than assumed.

3.1 Warehouse intake — the most urgent item on this document

This has two independent signals pointing the same direction: the voice note's detailed receiving workflow (§10–11), and the CEO's direct message asking for a Sortly review "today" specifically because "we have to replace this tool." Treat this as the priority, not a background Stage-2 line.

Scope, per voice note §10: open app → select "Receive Inventory" → optional photo → select product/category → select attributes → enter quantity → confirm. If the product doesn't exist, an authorised user creates it — this is exactly the controlled-vocabulary gate already designed in the spec's colour model (§6.4), so no new mechanism is needed, only a screen.

Explicitly not in this pass: AI-assisted photo recognition (§11) — the voice note itself calls this a "later" feature.

3.2 Automated low-stock alerting

The detection half of this is already built: `ThresholdsService.evaluate()` auto-drafts a restock request the moment stock crosses a threshold, deduplicated so a busy weekend cannot flood the queue. What Document 2's Stage 1 explicitly withheld was *surfacing* — requiring a human to check the dashboard rather than being told. Pulling this forward means the dashboard leads with what's flagged, prominently, by default.

Open question, not a blocker: proactive push/email/SMS delivery needs a channel decision (a mail or SMS provider) that hasn't been made. Detection and on-screen surfacing move to v1 now; off-screen delivery is scoped once a channel is chosen, so this isn't silently promised before it can be honestly built.

3.3 Restock quantity suggestion ("Smart Shipments", scoped to what v1 can support)

More of this already exists than the voice note's ask implies. `VelocitySeeder` already computes real weekly sell-through per item per location from the 2025 season data, and deliberately stops at style level — never colour — for exactly the reason Document 2 states: 2025 colour data covers 4.8% of revenue and building on it would be building on fantasy.

The voice note's own worked example (§20) is style-level too — "1,500 requested, 1,000–1,200 historically sold, recommend ~1,200" — no colour anywhere in it. So the v1 version is: when a request is raised, show the system-recommended quantity alongside the requested one, computed from that same velocity data plus current stock and remaining season days. This is surfacing existing data, not a new prediction engine.

Explicitly not in this pass: cross-market historical comparison, new-market cold-start modelling (§22), safety-buffer tuning beyond a simple fixed margin. Those need more history than one season

provides.

3.4 Overalllocation guard

When evaluating an open request, sum every other open request against the same warehouse stock and flag — not silently cap — if fulfilling this one in full would leave another location short. The recommended-vs-requested split from §3.3 is the natural place this surfaces.

3.5 Shipment consolidation

At pack time, show other open or soon-due requests for the same destination so a packer can combine boxes into one dispatch rather than two. A view on existing data, not a new subsystem.

3.6 Roles

Split the single `WAREHOUSE` role into `WAREHOUSE_MANAGER` (create shipments, adjust stock, approve requests) and `WAREHOUSE_OPERATOR` (receive, pack, cannot create products or touch the catalog) — mirroring the same shared-login problem the original Square audit found and flagging it as the reason. Add `SALES_OPERATOR` for staff who only need Square, never the app, so the smallest-privilege principle Document 2 already argued for actually holds for every role, not just some.

3.7 Dashboard status zones

Label existing threshold data as Healthy / Low / Critical / Out-of-Stock rather than a bare number. No new computation — a presentation layer on data already derived.

4. What softened

4.1 QR box tracking: mandatory → optional

Verified directly against the code before writing this: `dispatch()` already takes a box ID and posts the transfer to the ledger — it has never required a scan to run. The `/scan` screen is a convenience for finding that ID quickly, layered on top of the ledger transfer, not load-bearing underneath it. **No rework is needed.** The system already satisfies the voice note's own test in §14: shipment records, location inventory and Square sales deductions are the mechanism; scanning is optional confirmation on top, exactly as asked.

5. The one item that needs a fact, not a decision

5.1 Does any market physically operate a sub-warehouse today?

Document 2's "one warehouse" rule was a specific, verified observation about how this client's markets actually run — chaotic, staff grabbing stock freely, nothing counted on site. The voice note's three-tier idea (Chicago → market warehouse → booth) is explicitly hedged: *"not every market necessarily requires a separate warehouse, but the system should support this structure."*

Checked against the schema: this does not require a new data model. A location already has a `kind` of `MARKET` or `WAREHOUSE`, and a transfer already takes any two locations as endpoints — nothing about the ledger assumes exactly one warehouse exists. A market's local storage, if one is real, is simply a second `WAREHOUSE`-kind location, and Chicago → that location → the market's Square location is two ordinary transfers, not a new mechanism.

So this is designed-for already, not a build item. The only reason it isn't active today is that no second real warehouse has been confirmed to exist. Ask directly: does any market run a physical sub-warehouse right now, or is this the CEO future-proofing for growth? If real, it's a data entry — add the location, no code change. If aspirational, leave it exactly as designed and revisit when it's true.

6. What this document deliberately leaves alone

- Sept 12 catalog freeze, Sept 18 pilot target — unchanged
- The single catalog write window before the season, and every safety guard around it — unchanged
- No colour-level prediction, this season or next, until a full season of family-level sell-through exists to train on
- Write-off flow and season-close reconciliation — Stage 2, not mentioned in the voice note, no reason to move
- AI-assisted photo recognition — deferred by the voice note's own wording

Prepared by the build team, 24 Aug 2026, reconciling Document 2 against the CEO voice-note transcript per explicit instruction that the transcript carries higher authority on scope questions where the two disagree.